NAME: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 4 CHRONOLOGICAL ENQUIRY SEQUENCE • C1900–PRESENT

Medicine in Modern Britain, c1900–present

c1900–present: Ideas on Causes, Diagnostic Technology, The NHS, Penicillin & Lung Cancer



HISTORICAL PRIMARY RECORD • 5 JULY 1948 COI-INF-12-88

Aneurin Bevan with First NHS Patient Sylvia Beckingham, Park Hospital

Central Office of Information • Minister of Health Aneurin Bevan inaugurating the National Health Service at Park Hospital, Davyhulme, marking universal free access to healthcare.

PRIMARY VISUAL EVIDENCE THE NATIONAL ARCHIVES • INF 12/88

PEARSON EDEXCEL OPTION 11 • KEY TOPIC 4 CONTENT OFFICIAL SPECIFICATION DIRECTORY

1 Ideas About the Cause of Disease and Illness

- Advances in understanding the causes of illness and disease: the influence of genetic and lifestyle factors on health.
- Improvements in diagnosis: the impact of the availability of blood tests, scans and monitors.

2 Approaches to Prevention and Treatment

- The extent of change in care and treatment. The impact of the NHS and science and technology: improved access to care; advances in medicines, including magic bullets and antibiotics; high-tech medical and surgical treatment in hospitals.
- New approaches to prevention: mass vaccinations and government lifestyle campaigns.

3 Case Studies

- Key individuals: Fleming, Florey and Chain's development of penicillin.
- The fight against lung cancer in the twenty-first century: the use of science and technology in diagnosis and treatment; government action.

Living Timeline • Part 1: Magic Bullets, Blood Groups & Early Antibiotics (1900–1940)

Living Timeline Dual-Coding Protocol: Draw visual symbols for ABO agglutination, Salvarsan magic bullets, Fleming's mould petri dish, and Prontosil sulfa dye across the timeline below.

1901 • Karl Landsteiner: ABO Blood Groups Discovered

Key Topic 4.1

Landsteiner identifies A, B, and O blood groups, proving agglutination occurs when incompatible blood types mix. This discovery transforms blood transfusions from fatal gambles into safe, predictable surgical lifelines, foundational for modern trauma surgery.

1909 • Paul Ehrlich & Sahachiro Hata: Salvarsan 606

Key Topic 4.3

Building on dye research, Ehrlich and Hata systematically test hundreds of arsenic compounds, discovering arsphenamine (compound 606). Humanity's first synthetic chemical 'magic bullet' targets syphilis bacteria without poisoning human tissue.

1928 • Alexander Fleming: Accidental Discovery of Penicillin

Key Topic 4.4

Returning from holiday, Fleming notices a contaminated staphylococcus culture dish with a halo of destroyed bacteria around Penicillium notatum mould. Fleming publishes in 1929, though unable to purify or extract enough active substance for clinical trials.

1932 • Gerhard Domagk: Prontosil & Sulfonamide Drugs

Key Topic 4.3

Domagk demonstrates that the red leather dye Prontosil cures streptococcal blood poisoning in mice (and saves his own daughter's arm). Chemists isolate the active ingredient sulfonamide, launching the world's first mass-prescribed class of antibacterial wonder drugs.

Living Timeline Dual-Coding Protocol: Draw visual symbols for Peoria deep fermentation vats, the 1948 NHS charter, Franklin's Photograph 51 double helix, and CT scan slices across the timeline below.

1941–44 • Florey, Chain & Heatley: Penicillin Mass Production

Key Topic 4.4

Howard Florey and Ernst Chain purify penicillin at Oxford, successfully treating mice and Albert Alexander (1941). Heatley and Florey travel to Peoria, Illinois, using corn-steep liquor and deep-tank fermentation to produce 2.3 million doses for D-Day casualties.

1948 • Aneurin Bevan: Foundation of the NHS

Key Topic 4.3

On 5 July 1948, the National Health Service opens at Park Hospital, Manchester. Championed by Bevan and the 1942 Beveridge Report, it nationalises voluntary and municipal hospitals, providing universal healthcare free at the point of clinical delivery.

1953 • Watson, Crick & Franklin: DNA Double Helix Structure

Key Topic 4.1

Using Rosalind Franklin and Maurice Wilkins's X-ray crystallography (Photograph 51), Watson and Crick deduce the double-helix structure of DNA at Cambridge, unlocking hereditary genetic code and modern molecular gene therapy.

1971–2003 • CT Scanners (1971) & The Human Genome Project (2003)

Key Topic 4.2 • 4.1

Godfrey Hounsfield invents the EMI CT scanner, combining computer algorithms with X-rays to image internal organs without surgery. In 2003, the international Human Genome Project maps all 3 billion chemical base pairs in human DNA, founding precision medicine.

Why did the discovery of DNA revolutionize understanding of disease causes, yet take decades to yield treatments?

AUDIO • QUIZ

1 c1900–1940s • Why did doctors assume proteins carried hereditary traits?

- Gregor Mendel peas
- Chromosomes observed
- Hereditary protein theory

2 1951–1952 • How did Photograph 51 prove DNA's helical structure?

- Rosalind Franklin
- X-ray crystallography
- Photograph 51 (1952)

3 1953 • How did base pairing explain hereditary replication?

- Watson & Crick model
- Double helix structure
- DNA base pairing

4 1990–2003 • Why was mapping 3 billion base pairs a medical milestone?

- Human Genome Project
- Mapped 3bn base pairs
- International teamwork

5 2000s–Present • Why did understanding causes outpace finding cures?

- Genetic screening
- BRCA1/2 gene testing
- Gene therapy lag

Franklin's 62-hour X-ray exposure revealed the double helix; Watson and Crick celebrated in The Eagle pub.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why there was rapid progress in understanding the causes of disease in the period c1950 to the present. [12 marks]

Stimulus: **Rosalind Franklin's Photograph 51 (1952)** | **The Human Genome Project (1990–2003)** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.1). In the drawing box, sketch Franklin's Photograph 51 cross pattern and Watson & Crick's double helix model. Annotate how base pairing unlocked the causes of hereditary disease! **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which scientific methods for investigating disease causes in the late twentieth century (c1950–c2000) were different from methods used in the late nineteenth century (c1870–c1900). [4 marks]

Contrast computerized DNA sequencing with manual microscopic staining with chemical dyes.

Were high-tech diagnostic scanners more vital to modern health than state lifestyle campaigns?

AUDIO • QUIZ

1 1895–1918 • How did X-rays transform surgical trauma diagnosis in WWI?

- Wilhelm Röntgen (1895)
- Marie Curie mobile vans
- Shrapnel detection

2 1930s–1960s • Why was fiber-optic endoscopy superior to exploratory surgery?

- Blood tests & typing
- Centrifuges & biopsy
- Fiber-optic endoscopes

3 1971–1980s • Why did CT scans transform internal soft-tissue diagnosis?

- Godfrey Hounsfield (1971)
- EMI CT body scanning
- MRI magnetic imaging

4 c1950–1980s • What drove the shift from infectious to lifestyle diseases?

- Antibiotics curb fevers
- Higher life expectancy
- Cancers & heart disease

5 1980s–Present • Why do governments favour lifestyle prevention over treatment?

- 1983 Seatbelt legislation
- 2007 Public smoking ban
- Sugar tax & Change4Life

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'Technological innovations in medical diagnosis were more important than public health lifestyle campaigns in improving health in Britain c1900–present.' How far do you agree? [16+4 marks]

Stimulus: **Wilhelm Röntgen's X-rays (1895)** | **The 2007 smoking ban in public places** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *On the one hand, high-tech diagnosis was vital because... • For example, ... • This was significant because... • On the other hand, lifestyle campaigns mattered because... • In particular, ... • Overall, the most decisive factor was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.2). In the drawing box, sketch Bertha Röntgen's hand X-ray and Godfrey Hounsfield's circular CT scanner. Annotate how scanning permanently ended blind exploratory surgery! **[ ] Done**

Hounsfield's CT scanner combined rotating X-rays with digital computers to slice through soft body tissue.

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which methods of diagnosing illness in the late twentieth century (c1970–present) were different from methods used in the nineteenth century (c1800–c1900). [4 marks]

Contrast non-invasive 3D internal scanning and blood labs with external observation and exploratory surgery.

Was the establishment of the NHS in 1948 more transformative than the development of magic bullets?

AUDIO • QUIZ

1 1906–1909 • Why was Salvarsan 606 termed the first "magic bullet"?

- Paul Ehrlich research
- Magic bullet concept
- Salvarsan 606 (1909)

2 1932–1935 • How did Prontosil prove synthetic drugs could kill microbes?

- Gerhard Domagk (1932)
- Prontosil red dye
- Sulfa drug revolution

3 1942 • How did WWII unity create momentum for state healthcare?

- 1942 Beveridge Report
- Five Giant Evils
- Post-war welfare state

4 1945–1948 • How did Bevan compromise to persuade resistant doctors?

- Aneurin Bevan (1948)
- BMA doctor compromise
- Free universal healthcare

5 1950s–Present • Why has modern medical success created financial strain?

- Kidney dialysis units
- Hip replacement surgery
- Rising NHS cost pressures

Ehrlich tested 605 failed arsenic compounds before compound 606 finally cured syphilis in 1909.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'The establishment of the National Health Service in 1948 was the most significant turning point in medical care in the period c1900–present.' How far do you agree? [16+4 marks]

Stimulus: **The National Health Service (1948)** | **The discovery of Salvarsan 606 (1909)** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *On the one hand, the NHS was transformative because... • For example, ... • This was significant because... • On the other hand, magic bullets were crucial because... • In particular, ... • Overall, the most decisive factor was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.3). In the drawing box, sketch Ehrlich's test tube of Salvarsan 606 and the 1948 NHS information leaflet. Annotate how free healthcare abolished medical poverty! **[ ] Done**

Bevan overcame doctor opposition by "stuffing their mouths with gold," launching the NHS on 5 July 1948.

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which healthcare access in Britain after 1948 was different from healthcare access in the 1930s. [4 marks]

Contrast free universal healthcare funded by taxation with private fees and voluntary charity hospitals.

Why did penicillin require a global war and American industrial might to become a mass-produced cure?

AUDIO • QUIZ

1 1928–1929 · Why could Fleming not turn mould into a clinical medicine?

- Alexander Fleming (1928)
- Penicillium notatum mould
- Staphylococcus halo

2 1938–1940 · How did Norman Heatley solve the purification challenge?

- Florey & Chain (Oxford)
- Norman Heatley apparatus
- Freeze-drying extract

3 1940–1941 · Why was Alexander's trial both a success and a tragedy?

- 1940 Lethal mice trial
- Albert Alexander (1941)
- Supply exhaustion death

4 1941–1944 · How did US industrial fermentation enable mass production?

- Peoria, Illinois lab
- Corn-steep liquor vats
- 2.3m D-Day doses

5 1944–Present · Why did Fleming warn that antibiotic overuse causes resistance?

- Allied troop survival
- 1945 Nobel Prize
- MRSA superbug threat

Fleming almost washed his petri dish in Lysol, but noticed a clear halo where mould destroyed golden staphylococci.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why penicillin was successfully developed and mass-produced in the period 1928–1945. [12 marks]

Stimulus: **Alexander Fleming's discovery (1928)** | **The impact of the Second World War** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.4). In the drawing box, sketch Fleming's contaminated petri dish and an American 10,000-gallon deep-tank vat. Annotate how penicillin saved Allied troops on D-Day! **[ ] Done**

The Oxford team grew penicillin in bedpans; American factories brewed it in 10,000-gallon fermentation vats.

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]**

SCORE: [/ 4] • 5 MINS

Explain one way in which Alexander Fleming's discovery of penicillin in 1928 was similar to Edward Jenner's discovery of the smallpox vaccine in 1796. [4 marks]

Focus on both breakthroughs relying on accidental observations of natural organisms protecting against disease.

Why has the 21st-century fight against lung cancer required state coercion rather than high-tech cures?

AUDIO • QUIZ

1 2000s–Present • Why is early lung cancer so difficult to detect?

- Few pain receptors
- Asymptomatic growth
- Late Stage 3/4 detection

2 2000s–Present • How do CT and PET scans improve cancer detection?

- High-resolution CT scans
- PET-CT tracer imaging
- Bronchoscopy & biopsy

3 2000s–Present • How has technology reduced treatment trauma?

- VATS keyhole lobectomy
- Chest cracking avoided
- CyberKnife pinpoint rays

4 2000s–Present • How does immunotherapy harness the immune system?

- Tumour DNA sequencing
- Targeted pharmacotherapy
- Pembrolizumab T-cell boost

5 2006–Present • Why has state legislation proved more effective than cure?

- 2007 Public smoking ban
- 2016 Plain packaging
- Smoke-free generation law

Despite high-tech oncology, 10-year lung cancer survival is only ~10%: statutory prevention remains far more powerful than clinical cure.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'Government action was the main reason for progress in combating lung cancer in the twenty-first century.' How far do you agree? [16+4 marks]

Stimulus: **The Health Act 2006 (2007 public smoking ban)** | **High-tech diagnostic scans and treatments** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *On the one hand, government action was decisive because... • For example, ... • This was significant because... • On the other hand, high-tech treatments mattered because... • In particular, ... • Overall, the most decisive factor was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.5). In the drawing box, sketch a 21st-century CT scanner and a 2016 olive-green plain pack. Annotate how state legislation and high-tech scans combat lung cancer! **[ ] Done**

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which the diagnosis of lung cancer was different in the twenty-first century from diagnosis in the nineteenth century. [4 marks]

Focus on nineteenth-century doctors relying on external stethoscopes and post-mortem autopsies, whereas twenty-first-century clinicians use high-resolution CT scans and bronchoscopy to detect living tumors.

KEY TOPIC 4 NOTES & EXAMINATION STRATEGY

STUDENT ASSESSMENT RECORD • KEY TOPIC 4 TRACKER

Paper 1: Medicine in Britain, c1250–present • Modern Britain (c1900–present)

PUPIL: _____ **TARGET GRADE:** ☐ **PREDICTED:** ☐ **ATTITUDE:** 1 • 2 • 3 • 4 • 5

#	ENQUIRY / SPECIFICATION FOCUS	DO NOW (10M)	Q3 SIM/DIFF (4M)	Q4/5 EXTENDED	LESSON TOTAL
1	**KT4.1: Ideas on Causes: Genetics, DNA & The Human Genome Project** *"Why did the discovery of DNA revolutionize understanding of disease causes, yet take decades to yield treatments?"*	/ 10	/ 4	Q4 Explain: / 12	/ 26
2	**KT4.2: Lifestyle Factors & The Technological Revolution in Diagnosis** *"Were high-tech diagnostic scanners more vital to modern health than state lifestyle campaigns?"*	/ 10	/ 4	Q4 Explain: / 12	/ 26
3	**KT4.3: Magic Bullets, High-Tech Treatments & The Birth of the NHS** *"Was the establishment of the NHS in 1948 more transformative than the development of magic bullets?"*	/ 10	/ 4	Q4 Explain: / 12	/ 26
4	**KT4.4: Case Study 1: The Antibiotic Revolution: Fleming, Florey & Chain and Penicillin** *"Why did penicillin require a global war and American industrial might to become a mass-produced cure?"*	/ 10	/ 4	Q4 Explain: / 12	/ 26
5	**KT4.5: Case Study 2: The Fight Against Lung Cancer (21st Century)** *"Why has the 21st-century fight against lung cancer required state coercion rather than high-tech cures?"*	/ 10	/ 4	Q5/6 Essay: / 20	/ 34
KEY TOPIC 4 CUMULATIVE ASSESSMENT TOTALS		**Do Now: / 50**	**Q3: / 20**	**Ext: / 68**	**/ 138**

TEACHER FORMATIVE ASSESSMENT • WWW / EBI FEEDBACK

WHAT WENT WELL (WWW):

_____

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_____

EVEN BETTER IF (EBI):

_____

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Teacher Signature: _____ **Date:** _____

DIGITAL QUIZZING HUB • 20-QUESTION PRACTICE QUIZZES

Scan with smartphone camera to open live interactive 20-question self-marking quizzes

KT 4.1	KT 4.2	KT 4.3	KT 4.4	KT 4.5
SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ
Score: / 20	**Score: / 20**	**Score: / 20**	**Score: / 20**	**Score: / 20**

Retain this student assessment record for parent consultations and GCSE revision monitoring.